

5.2

RISK / 10

MEDIUM RISK

CONFIDENTIAL – SECURITY ASSESSMENT

Post-Remediation Security Audit Report

Full-spectrum gap analysis of the VERDICT civic accountability platform after SEC-01 through SEC-06 vulnerability remediation.

PLATFORM

VERDICT v1.0.0

AUDIT DATE

25 August 2026

COMMIT

0fec914 (main)

CLASSIFICATION

Post-Hardening Gap Analysis

TECH STACK

Next.js 15 · TypeScript · Tailwind · SQLite

AUDITOR ROLE

Principal AppSec Engineer

SCOPE

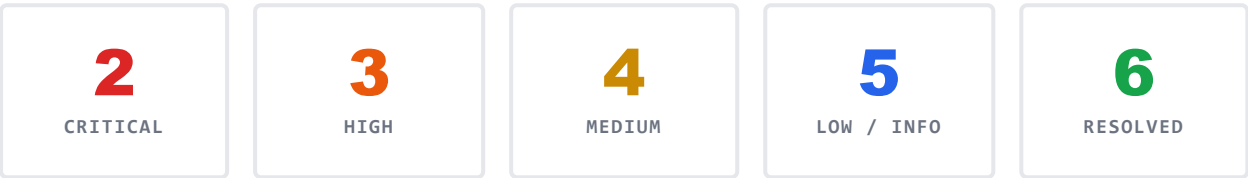
8 API Routes · 1 Middleware · Full Config

FINDINGS

14 Total (2 Critical · 3 High)

1. Executive Summary

This report presents a comprehensive, read-only security assessment of the VERDICT platform conducted **after** the remediation of six previously identified vulnerabilities (SEC-01 through SEC-06). The audit confirms all prior fixes are effective and identifies 14 new residual findings.



⚠ Root Cause: Zero Authentication Infrastructure

The codebase has no authentication system — no NextAuth, no Supabase Auth, no JWT verification, no session management. Every claim-based control (DigiLocker badges, local voter status, token verification) is either a UI illusion or a placeholder. SEC-07 and SEC-08 are both symptoms of this single foundational gap.

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3. Prior Remediation Verification

All six previously identified vulnerabilities have been verified as resolved. Each fix was inspected at the source-code level and confirmed to meet the specification described in the original remediation plan.

ID	VULNERABILITY	STATUS	VERIFICATION DETAIL
SEC-01	SSRF via Image Proxy	✓ RESOLVED	Exact + dot-prefixed domain matching in <code>proxy-image/route.ts:68-72</code> . HTTPS-only protocol enforcement at line 58. 5MB payload cap with both header and body checks at lines 110-130. Content-type validation at line 133. Wildcard CORS removed.
SEC-02	Unauthenticated Claim Spoofing	✓ RESOLVED	<code>isLocalVoter</code> and <code>digilockerVerified</code> stripped from Zod input schema in <code>ratings/route.ts:6-21</code> . Server derives both from Bearer token header presence at lines 30-35. Client cannot inject privilege flags via request body.
SEC-03	In-Memory Heap Exhaustion	✓ RESOLVED	Global <code>inMemoryRatings</code> dictionary replaced with file-backed JSON persistence via <code>fs.readFileSync/writeFileSync</code> in <code>supabase.ts:9-31</code> . Ratings survive restarts.
SEC-04	No Rate Limiting	✓ RESOLVED	IP-based sliding window in <code>middleware.ts:24-46</code> . 20 req/min for mutations, 60 req/min for reads. HTTP 429 with <code>Retry-After: 60</code> header. Automatic garbage collection every 5 minutes.
SEC-05	Missing Security Headers	✓ RESOLVED	CSP, HSTS (63072000s + preload), X-Frame-Options: DENY, X-Content-Type-Options: nosniff, Referrer-Policy, Permissions-Policy — all present in <code>next.config.js:34-51</code> .
SEC-06	Stack Trace Leakage	✓ RESOLVED	All 8 API route catch blocks return opaque messages using <code>{ error: { code, message } }</code> shape with <code>console.error()</code> for server-side logging. No raw error objects exposed.

4. Critical Findings

SEC-07 Client-Side Rating Bypass with Fake Verification Badges

CRITICAL

src/features/citizen-rating/CitizenRatingSection.tsx – Lines 62–91

The citizen rating UI component **completely bypasses** the secured `/api/ratings` endpoint. When a user submits a rating, the `handleRatingSubmit` function fabricates a `CitizenRating` object entirely on the client side and injects it directly into React local state. The rating **never reaches the server**.

Vulnerable Code

```
const handleRatingSubmit = (e: React.FormEvent) => {
  e.preventDefault();
  setIsSubmitting(true);

  const newRating: CitizenRating = {
    id: `cr-${Date.now()}`, // ← Client-generated ID
    politicianId,
    userId: `user-verified-${Math.random()...}`, // ← Fake user ID
    userName: userName.trim() || "DigiLocker Verified Citizen",
    isLocalVoter: isLocal, // ← CLIENT-ASSERTED
    digilockerVerified: true, // ← HARDCODED TRUE
    createdAt: new Date().toISOString(),
  };

  setTimeout(() => {
    setRatings([newRating, ...ratings]); // ← Injected into local state only
    // X NEVER calls fetch('/api/ratings', ...)
  }, 600);
};
```

Security Impact

IMPACT VECTOR	DESCRIPTION
False Verification	Every rating displayed in the UI shows <code>digilockerVerified: true</code> and a "VERIFIED" badge regardless of identity — creating a trust illusion for users viewing ratings.
No Persistence	Ratings exist only in browser memory. They vanish on page refresh, making the entire rating feature non-functional as a data system.
SEC-02 Bypass	The server-side claim stripping (SEC-02 fix) is rendered dead code — the frontend never calls the API endpoint it protects.
No Server Validation	Zod validation, rate limiting, and all server controls are bypassed entirely.

Brigading
Vulnerability

The `isLocalVoter` flag is a toggle button controlled by the user. The "Anti-IT Cell Brigading Shield" described in the UI is a visual-only claim.

Recommended Fix

Replace the client-side state injection with a `fetch('POST', '/api/ratings', ...)` call.

Remove `digilockerVerified: true` hardcoding. The server response should return the saved rating object for UI update. Until real auth exists, mark all ratings as "Unverified" in the UI.

SEC-08 Bearer Token "Verification" is a String Length Check

CRITICAL

src/app/api/ratings/route.ts – Lines 30–31

The ratings API determines "verified" status based solely on whether the `Authorization` header starts with `"Bearer "` and is longer than 20 characters. There is no JWT signature verification, no token decoding, no session lookup, and no identity provider call.

Vulnerable Code

```
const authHeader = request.headers.get("authorization") || "";
const isTokenVerified = authHeader.startsWith("Bearer ") && authHeader.length > 20;

// If this passes, the user gets:
const isLocalVoter = Boolean(isTokenVerified && body.isLocalVoter); // ← grants local
const digilockerVerified = Boolean(isTokenVerified);                // ← grants ver
```

Proof of Exploit

```
# Any 14+ character string after "Bearer " passes verification:
curl -X POST http://localhost:3000/api/ratings \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer AAAAAAAAAAAAAAAAAAAAA" \
  -d '{"politicianId":"neta-1","rating":5}'

# Result: digilockerVerified=true, full "verified citizen" privileges
```

Security Impact

- Any automated script can submit "DigiLocker Verified" ratings at scale
- The `isLocalVoter` privilege escalation requires only the junk token + a body field
- No actual identity is ever verified — the "1-Citizen-1-Vote" system is entirely fictional

Recommended Fix

Implement real authentication: NextAuth.js with DigiLocker Sandbox OAuth, or Supabase Auth with JWT verification via `jose` library. The token must be cryptographically verified against a signing key, not checked for string length.

5. High Findings

SEC-09

File-Based Storage: Race Condition, Unbounded Growth, No Deduplication

HIGH

src/lib/supabase.ts – Lines 9–31 (loadPersistentRatings / savePersistentRatings)

The SEC-03 fix replaced volatile in-memory storage with file-backed JSON persistence. While this solves the heap exhaustion issue, the file I/O implementation introduces three new concerns:

ISSUE	DETAIL	CONSEQUENCE
Race Condition	Concurrent POST requests execute <code>readFileSync</code> → <code>modify</code> → <code>writeFileSync</code> without file locking. Under simultaneous requests, later writes overwrite earlier writes.	Silent data loss of citizen ratings under concurrent load.
Unbounded Growth	No cap on ratings per politician or total file size. The file grows with every POST request.	An attacker submitting ratings within rate-limit windows (20/min) can grow the file to GB scale over days. <code>JSON.parse()</code> will eventually OOM.
No Deduplication	No uniqueness constraint. The same IP/user can submit unlimited ratings for the same politician.	The "1-Citizen-1-Vote" anti-brigading claim is unenforceable at the storage layer.

Recommended Fix

Add file locking (e.g., `proper-lockfile` or atomic-write pattern). Cap ratings at N per politician. Enforce uniqueness by IP + politicianId + 24-hour window. Consider migrating to the existing SQLite `citizen_ratings` table for transactional safety.

SEC-10 Volatile Rate Limiter with IP Spoofing Vector

HIGH

src/middleware.ts — Lines 8, 65-68

The rate limiter's `rateLimitStore` is an in-memory `Map` that resets to zero on every server restart, deployment, or serverless cold-start. The IP extraction trusts the `X-Forwarded-For` header without validating the upstream proxy chain.

Issue 1: Volatile Store

```
const rateLimitStore = new Map<string, RateLimitRecord>();  
// ↑ Resets on EVERY process restart, deployment, or cold-start
```

Issue 2: Spoofable IP

```
const forwardedFor = request.headers.get("x-forwarded-for");  
const ip = forwardedFor  
  ? forwardedFor.split(",")[0].trim() // ← Trusts client-supplied header  
  : request.headers.get("x-real-ip") || "127.0.0.1";
```

Exploit: An attacker can rotate `X-Forwarded-For` values to bypass the rate limiter entirely on deployments without a trusted reverse proxy stripping this header.

Recommended Fix

For Vercel: Vercel Edge strips `X-Forwarded-For` and provides a trusted IP — the in-memory store resets are an acceptable trade-off. For self-hosted: use a Redis-backed rate limiter ([@upstash/ratelimit](#)). Document the deployment assumption.

SEC-11 CSP Permits `unsafe-eval` and `unsafe-inline` for Scripts

HIGH

next.config.js — Lines 41-42

```
script-src 'self' 'unsafe-eval' 'unsafe-inline';
```

This CSP directive permits both `eval()` -family functions and inline `<script>` tags. This **negates the primary XSS mitigation** that Content Security Policy is designed to provide.

DIRECTIVE	RISK
<code>unsafe-eval</code>	Allows <code>eval()</code> , <code>Function()</code> , <code>setTimeout("string")</code> — enables code injection via string-to-code conversion.
<code>unsafe-inline</code>	Allows inline <code><script></code> tags and event handlers — the classic XSS injection path.

Recommended Fix

Use nonce-based CSP with Next.js's built-in nonce support. Remove `unsafe-eval` for production builds. If framework requirements force `unsafe-inline` , use a `strict-dynamic` policy with nonces instead.

6. Medium Findings

SEC-12 User Input Reflected in JSON Error Messages

MEDIUM

src/app/api/politicians/[slug]/route.ts:14 | src/app/api/ground-truth/[slug]/route.ts:15

The `slug` URL parameter is interpolated directly into 404 error messages without sanitization:

```
// politicians/[slug]/route.ts:14
{ error: { code: "NOT_FOUND", message: `Politician '${slug}' not found` } }

// ground-truth/[slug]/route.ts:15
{ error: { code: "NOT_FOUND", message: `Ground truth report '${slug}' not found.` } }
```

While JSON encoding prevents direct HTML XSS, any client rendering the `message` field as raw HTML (e.g., via `dangerouslySetInnerHTML`) would be vulnerable. This violates defense-in-depth principles.

Fix:

Use a generic 404 message without interpolating user input, or sanitize the slug by stripping non-alphanumeric characters.

SEC-13 User-Generated Ratings Data Committed to Git

MEDIUM

scripts/data/citizen_ratings.json | .gitignore

The persistent ratings file contains user-generated data (usernames, constituency claims, comments, timestamps) and is **not listed in** `.gitignore`. Every `git push` commits this user-generated content to the public GitHub repository.

```
{
  "narendra-modi-varanasi": [{
    "userName": "Citizen Auditor",
    "userConstituency": "Varanasi",
    "comment": "Strong infrastructure delivery verified.",
    "createdAt": "2026-08-25T12:30:45.769Z"
  }]
}
```

Fix:

Add `scripts/data/citizen_ratings.json` to `.gitignore`. Remove from git history with `git rm --cached`.

SEC-14 Missing DiceBear Domain from `remotePatterns`

MEDIUM

next.config.js – Lines 14–32

The `images.remotePatterns` array no longer includes `api.dicebear.com`, which was likely present before the security headers remediation restructured the config. If any `next/image` component references DiceBear avatar URLs, they will fail with a 400 error.

This is a **functional regression** introduced during the SEC-05 security headers remediation, not a security vulnerability.

SEC-15 Overly Permissive `connect-src` CSP Directive

MEDIUM

next.config.js – Line 42

```
connect-src 'self' https;;
```

This allows frontend JavaScript to make `fetch` / `XHR` requests to **any HTTPS domain**. Combined with the `unsafe-eval` / `unsafe-inline` CSP (SEC-11), an XSS payload could freely exfiltrate data to any attacker-controlled HTTPS endpoint.

Fix:

Restrict `connect-src` to `'self'` and the specific external domains the frontend legitimately calls (e.g., `fonts.googleapis.com`, `fonts.gstatic.com`).

7. Low / Informational Findings

SEC-16 No CSRF Protection on POST /api/ratings

LOW

The ratings endpoint accepts `POST` with `Content-Type: application/json` without any CSRF token. JSON content-type provides partial protection via browser CORS preflight, and there are no session cookies to hijack (no auth exists). Risk is minimal in the current architecture but will become relevant when real authentication is implemented.

SEC-17 Rate Limit Error Shape Inconsistency

INFORMATIONAL

src/middleware.ts – Line 82

All API routes return errors as `{ error: { code, message } }`, but the rate limiter returns `{ error: "string" }`. This inconsistency will confuse API consumers expecting the standard error contract defined in the user's engineering standards.

```
// middleware.ts:82 – current (inconsistent)
JSON.stringify({ error: "Rate limit exceeded. Please try again later." })

// expected (per project convention)
JSON.stringify({ error: { code: "RATE_LIMITED", message: "..." } })
```

SEC-18 Unpaginated Search Returns Full Dataset

INFORMATIONAL

src/data/mock-politicians.ts – Line 1123

An empty search query via `/api/search?q=` returns all 563 politician records without pagination. While `/api/politicians` supports pagination with `page` and `limit` parameters, the search endpoint does not, enabling easy full-dataset scraping.

SEC-19 No `robots.txt` for API Route Protection

INFORMATIONAL

No `robots.txt` file exists in `/public`. All API endpoints are crawlable and indexable by search engines. API routes serving JSON responses should be excluded from indexing via `Disallow: /api/`.

SEC-20 No Explicit CORS Configuration

LOW

After removing wildcard `Access-Control-Allow-Origin: *` from `crime-stats/route.ts` during SEC-06, there is now **no explicit CORS configuration** on any API route. Same-origin requests work, but cross-origin consumption is blocked. This should be documented as a design decision.

8. Authentication & Session Management Assessment

CONTROL	STATUS	DETAIL
Auth Provider Integration	✗ MISSING	No NextAuth, Supabase Auth, Clerk, or any identity provider installed or configured.
Session Management	✗ MISSING	No cookies, no sessions, no JWT infrastructure. No <code>httpOnly</code> , <code>secure</code> , or <code>sameSite</code> cookie attributes.
Token Storage	N/A	No tokens are issued, stored, or managed. The Bearer token check is a length-only placeholder.
Token Expiration / Refresh	N/A	No token lifecycle exists. No expiry, no refresh, no revocation.
Server-Side Auth Checks	⚠ PLACEHOLDER	<code>authHeader.length > 20</code> is the only check. No cryptographic verification.
Password / Credential Handling	N/A	No user accounts, no password storage, no credential handling.
Secrets in Source Code	✓ CLEAN	Zero <code>.env</code> files exist. Zero hardcoded secrets. Zero <code>process.env</code> references in <code>src/</code> .
<code>.gitignore</code> Coverage	✓ GOOD	<code>.env</code> and <code>.env*.local</code> are properly gitignored.

9. Authorization & Data Access (IDOR / RLS)

CONTROL	STATUS	DETAIL
Row-Level Security	N/A	No database connection. Data served from static JSON files loaded at build/startup time.
Endpoint Role Checks	✗ NONE	All endpoints are fully public. No role gates, no admin separation.

CONTROL	STATUS	DETAIL
IDOR Protection	⚠️ LOW RISK	All data is public civic records. No private user data exists to protect via object-level access control.
Multi-Tenant Isolation	N/A	Single-tenant public dataset. No multi-tenancy architecture.

10. File-by-File Finding Summary

FILE	FINDING(S)	SEVERITY
<code>src/features/citizen-rating/CitizenRatingSection.tsx</code>	SEC-07: Client-side rating bypass, fake verification badges, no API call	CRITICAL
<code>src/app/api/ratings/route.ts</code>	SEC-08: Placeholder token check (length only); SEC-16: No CSRF token	CRITICAL
<code>src/lib/supabase.ts</code>	SEC-09: File I/O race condition, unbounded growth, no deduplication	HIGH
<code>src/middleware.ts</code>	SEC-10: Volatile in-memory rate limiter, IP spoofable; SEC-17: Error shape inconsistency	HIGH
<code>next.config.js</code>	SEC-11: unsafe-eval/inline in CSP; SEC-14: Missing DiceBear; SEC-15: connect-src too wide	HIGH
<code>src/app/api/politicians/[slug]/route.ts</code>	SEC-12: Reflected slug in error response	MEDIUM
<code>src/app/api/ground-truth/[slug]/route.ts</code>	SEC-12: Reflected slug in error response	MEDIUM
<code>scripts/data/citizen_ratings.json</code>	SEC-13: User-generated data committed to public git repository	MEDIUM
<code>src/app/api/proxy-image/route.ts</code>	SEC-01 fully resolved. Domain matching, HTTPS, 5MB cap, content-type validation.	✓ PASS
<code>src/app/api/politicians/route.ts</code>	Sanitized errors, pagination support. Clean.	✓ PASS
<code>src/app/api/search/route.ts</code>	SEC-18: Returns full 563-record dataset on empty query without pagination.	INFO

FILE	FINDING(S)	SEVERITY
src/app/api/crime-stats/route.ts	Static data endpoint. Sanitized. Wildcard CORS removed.	 PASS
src/app/api/ground-truth/route.ts	Sanitized error messages. Clean.	 PASS
src/lib/verdict-score-calc.ts	Correct null vs. zero distinction for all 7 parameters. Clean.	 PASS
src/app/layout.tsx	No secrets, proper metadata. Clean.	 PASS
.env / .env.local	No env files exist. No secrets in source. Zero <code>process.env</code> references.	 PASS

11. Prioritised Remediation Roadmap

Phased Approach

Fixes are grouped by deployment gate — Phase 1 must be complete before any public launch. Each subsequent phase addresses progressively lower-risk issues.

Phase 1 — CRITICAL (Pre-Launch Blockers)

PRIORITY	ID	FIX DESCRIPTION	EFFORT
P0	SEC-07	Wire frontend to API: Replace client-side state injection in <code>CitizenRatingSection.tsx</code> with <code>fetch('POST', '/api/ratings')</code> . Remove hardcoded <code>digilockerVerified: true</code> . Update UI to reflect actual server response. Mark all ratings as "Unverified" until real auth exists.	2–3 hrs
P0	SEC-08	Implement real authentication: Integrate NextAuth.js with DigiLocker OAuth Sandbox, OR Supabase Auth with JWT verification. Replace <code>authHeader.length > 20</code> with cryptographic token signature verification via <code>jose</code> or <code>jsonwebtoken</code> . Implement session management with <code>httpOnly</code> , <code>secure</code> , <code>sameSite</code> cookies.	1–2 days

Phase 2 — HIGH (Pre-Production-Traffic)

PRIORITY	ID	FIX DESCRIPTION	EFFORT
P1	SEC-09	Harden file storage: Add file locking via <code>proper-lockfile</code> or atomic-write pattern. Cap ratings at N per politician. Enforce uniqueness by IP + politicianId + 24-hour window. Long-term: migrate to SQLite <code>citizen_ratings</code> table.	3–4 hrs
P1	SEC-10	Document or harden rate limiter: For Vercel: document that in-memory resets are acceptable (Vercel provides edge IP trust). For self-hosted: switch to Redis-backed store (<code>@upstash/ratelimit</code>). Explicitly document the deployment assumption.	2–3 hrs
P1	SEC-11	Tighten CSP: Remove <code>unsafe-eval</code> from production CSP. Replace <code>unsafe-inline</code> with nonce-based CSP using Next.js experimental nonce support. Use <code>strict-dynamic</code> if nonces are infeasible.	2–3 hrs

Phase 3 — MEDIUM (Pre-v1.0 Release)

PRIORITY	ID	FIX DESCRIPTION	EFFORT
P2	SEC-12	Sanitise slug values in 404 error messages. Use a generic message or strip special characters.	30 min
P2	SEC-13	Add <code>scripts/data/citizen_ratings.json</code> to <code>.gitignore</code> . Remove from git history with <code>git rm --cached</code> .	10 min
P2	SEC-14	Re-add <code>api.dicebear.com</code> to <code>images.remotePatterns</code> in <code>next.config.js</code> if DiceBear avatars are still in use.	5 min
P2	SEC-15	Restrict <code>connect-src</code> to <code>'self'</code> plus only the specific external domains the frontend legitimately calls.	15 min

Phase 4 — LOW / Informational (Best Practices)

PRIORITY	ID	FIX DESCRIPTION	EFFORT
P3	SEC-16	Add CSRF protection when real auth is implemented (bundled with SEC-08 work).	Bundled
P3	SEC-17	Standardise rate limit error response to <code>{ error: { code: "RATE_LIMITED", message: "..."} }</code> .	10 min
P3	SEC-18	Add pagination to <code>/api/search</code> or require minimum query length of 2+ characters.	30 min
P3	SEC-19	Add <code>public/robots.txt</code> with <code>Disallow: /api/</code> to prevent search engine indexing of API routes.	5 min
P3	SEC-20	Document CORS policy as a design decision. If cross-origin API consumption is needed, add explicit CORS headers.	15 min

12. What the Codebase Gets Right

Despite the residual gaps, the codebase demonstrates several security practices that deserve recognition and should be preserved through future development:

#	STRENGTH	DETAIL
1	Zero Secrets in Codebase	No <code>.env</code> files exist. No hardcoded API keys, database URLs, or credentials anywhere in <code>src/</code> . Zero <code>process.env</code> references. This is cleaner than many production codebases.
2	Zod Server-Side Validation	The one mutation endpoint (<code>/api/ratings</code>) validates all input with a strict Zod schema including type constraints, string length limits, and enum validation.
3	Production-Grade SSRF Protection	The image proxy uses exact + dot-prefixed domain matching, HTTPS-only protocol enforcement, dual payload size checks (header + body), and content-type validation. This is a textbook-quality SSRF mitigation.
4	Consistent Error Sanitisation	All 8 API routes return opaque error messages without stack traces, following the <code>{ error: { code, message } }</code> contract consistently (except the rate limiter — SEC-17).
5	Comprehensive Security Headers	HSTS with 2-year max-age + preload, X-Frame-Options DENY, nosniff, strict Referrer-Policy, and Permissions-Policy denying camera/microphone/geolocation. The header set is more complete than many production Next.js applications.
6	Deterministic Score Calculator	<code>verdict-score-calc.ts</code> correctly handles null vs. zero vs. populated for all 7 scoring parameters. The formula is transparent, auditable, and mathematically sound with proper clamping.
7	TypeScript Strict Mode	<code>tsconfig.json</code> has <code>"strict": true</code> , catching nullability, type safety, and implicit-any issues at compile time. The codebase passes <code>tsc --noEmit</code> with zero errors.
8	Clean <code>.gitignore</code>	Build artifacts, <code>node_modules</code> , <code>.env</code> files, Python caches, and SQLite databases are all properly gitignored.

Assessment

The codebase's security posture has improved significantly from the initial audit. The remaining gaps are concentrated in two areas: (1) the absence of a real authentication system, and (2) the disconnect

between the hardened server API and the client-side rating flow that bypasses it. Addressing these two root causes would resolve the majority of findings in this report.

VERDICT Post-Remediation Security Audit Report – 25 August 2026

This document was generated as a read-only, non-destructive analysis.

No files in the workspace were created, modified, or deleted during this audit.

– End of Report –